Jinbo Zhang

jinboz1@uci.edu
powerofjinbo.github.io | github.com/powerofjinbo

EDUCATION

University of California, Irvine — **Sep. 2023 – Jun. 2027 (expected)**
B.S. in Physics with Honors · GPA 3.81/4.00 — *Irvine, CA*

- **Graduate coursework:** Quantum Field Theory, Group Theory, Machine Learning & Statistics.

SELECTED PUBLICATIONS & RESEARCH MANUSCRIPTS

M. H. Fieg, P. J. Fox, **J. Zhang**, A. Ghosh, V. Varada, D. Whiteson. **Time-Dependent Signals of New Physics at the LHC.** arXiv:2605.11071. *Under review at Physical Review D (revised manuscript).*

J. Zhang, A. Frankenthal. **Canto: A Graph Neural Network Tracker for Displaced Particles.** *In final polishing for submission to Physical Review D.*

J. Zhang, B. Echenard, D. G. Hitlin. **A Muon Missing-Momentum Experiment.** [PDF] *LDMX Collaboration internal note; Caltech SURF final report.*

J. Zhang, J. Eysermans. **Measurement of H $\to$ WW* in the $q\bar{q}\,\ell\nu\,q\bar{q}'$ Final State at FCC-ee.** [PDF] *Manuscript complete; to be released as an FCC-ee note.*

OTHER WORK

Euclid Collaboration: X. Xu, R. Chen, T. Li, A. R. Cooray, . . . , **J. Zhang**, et al. **Euclid Quick Data Release (Q1). AgileLens: A Scalable CNN-Based Pipeline for Strong Gravitational Lens Identification.** *arXiv:2604.06648.*

C. M. Gonzalez, Y.-H. Chen, C. Effarah, **J. Zhang**, et al. **Gate-Tunable Two-Dimensional ITO Epsilon-Near-Zero Materials Fabricated by Liquid Metal Printing.** *Proc. SPIE PC13579, PC135791B (2025).*

H. W. Lee, Q. Dang, D. Dang, et al., incl. **J. Zhang**. **Active and Nonlinear Epsilon-Near-Zero Photonics.** *Proc. SPIE PC13892 (2026).*

TALKS & POSTERS

- "A Muon Missing-Momentum Experiment." **Invited talk**, LDMX Collaboration meeting (Oct. 19, 2026).
- "A Muon Missing-Momentum Experiment." Final research talk, Caltech SURF Program (Aug. 20, 2026).
- "Time-Dependent Signals of New Physics at the LHC." Poster, UCI Undergraduate Research Symposium (May 2026).
- "Liquid Metal Printed 2D ITO for Nanophotonic Applications" (with Y.-H. Chen et al.). Poster, UCI symposium (2025).

RESEARCH EXPERIENCE

Canto: GNN Tracking of Displaced Particles | Honors Thesis — **Jul. 2026 – Present**
Mentor: Prof. Andre Frankenthal — *University of California, Irvine*

- Diagnosed why GNN trackers trained on Standard Model events lose displaced tracks: in the Exa.TrkX pipeline the learned graph misses displaced hit pairs, the edge classifier learns that tracks start near the beam line, and connected components turn single edge errors into lost tracks.
- Designed and built **Canto**, which rebuilds each stage on the helix geometry of a track (learned helix-triplet graph, new edge classifier, Mutex-Watershed track builder with learned merging). Trained only on $t\bar{t}$ events, it finds **99.1% of displaced heavy-neutral-lepton daughters**, 1.1 points more than the signal-trained baseline, at a lower fake rate. **First author** of the resulting paper.
- Ran a preregistered study (seven claims, all hold): on new particles simulated (Pythia8, ACTS) after the design freeze Canto leads the strongest baseline by 2.3 points; with 200 pileup collisions (TrackML) it is at least as efficient as published GNN trackers.

Muon Missing-Momentum Search for Dark Matter | Caltech SURF — **Jun. 22 – Aug. 28, 2026**
Mentors: Prof. Bertrand Echenard and Prof. David Hitlin — *California Institute of Technology*

- Built in eight weeks the full simulation and analysis chain for an LDMX-style muon-beam missing-momentum experiment: a Geant4 model (40-layer LYSO active target with per-crystal readout, EM and hadronic calorimeters) and a 10^8-muon background campaign on a Slurm cluster. **First author** of the resulting manuscript.
- Mapped rare backgrounds at the 10^{-8} level: identified muon bremsstrahlung with photonuclear interactions and hard muon-nucleus scattering with escaping neutrons as irreducible, and traced each surviving topology to its cause.
- Designed a 625-node graph neural network over the full detector and benchmarked it against calorimetric vetoes, BDTs, and transformers, reducing the background to a mean of 1.6 events at 90% signal efficiency, with **zero background at** 10^8

muons in four of five trainings at a tightened working point, under a preregistered model-selection and validation protocol.

Precision Higgs Physics at FCC-ee: $\sigma(\mathrm{ZH}) \times \mathcal{B}(\mathrm{H} \to \mathrm{WW}^*)$ **Jan. 2026 – Sep. 2026**

Mentor: Prof. Christoph Paus *Massachusetts Institute of Technology*

- **Conceived and led, as first author**, a complete projected measurement of $\sigma(\mathrm{ZH}) \times \mathcal{B}(\mathrm{H} \to \mathrm{WW}^*)$ in the semileptonic $\mathrm{q\bar{q}}\,\ell\nu\,\mathrm{q\bar{q}}'$ final state at $\sqrt{s} = 240$ GeV (10.8 ab^{-1}, IDEA detector), owning the chain from event selection to final fit over 10^9 fully simulated events (32 processes, FCCAnalyses).
- Achieved a **sub-percent projected precision on the signal strength** ($\delta\mu/\mu = 0.545\%$) from a simultaneous two-category profile-likelihood fit in pyhf; validated it with Asimov closure tests, likelihood scans, cross-validation, and systematic selection scans.
- Developed a Z-priority jet-pairing algorithm that uses the on-shell Z to resolve the off-shell W^* topology, separated on- and off-shell leptonic-W topologies into two categories, and trained an XGBoost classifier per category on 26 observables (holdout AUC = 0.987, 0.986). [analysis code]

Time-Dependent Searches for New Physics at the LHC **May 2025 – May 2026**

Mentor: Prof. Daniel Whiteson *University of California, Irvine*

- Co-developed a resonance search that adds event time as an analysis dimension, opening sensitivity to time-modulated signals invisible to mass-only searches. **Co-authored the resulting paper, now under review at *Physical Review D*.**
- Extended CATHODE weakly supervised anomaly detection from one to two dimensions, learning background distributions jointly in invariant-mass and time sidebands for fully data-driven searches.
- Built the simulation-to-limits pipeline (MadGraph5, Pythia8, Delphes, classifier training, sideband-closure diagnostics, expected limits) and the mass-only vs. mass-plus-time sensitivity comparisons of the paper. [code]

ADDITIONAL RESEARCH

- **High-redshift galaxy evolution** (Prof. A. Cooray, UCI, 2024–25): Spitzer/ALMA photometry and SED fitting of dusty star-forming galaxies.
- **Epsilon-near-zero nanophotonics** (Prof. H. Lee, UCI, 2024–25): liquid-metal-printed, gate-tunable 2D ITO films and their nonlinear optics.

TECHNICAL SKILLS

HEP simulation & analysis	Geant4, ACTS, MadGraph5, Pythia8, Delphes, FCCAnalyses, ROOT/RDataFrame, pyhf
Machine learning	PyTorch, XGBoost, GNN tracking (Exa.TrkX), transformers, anomaly detection (CATHODE)
Programming & computing	Python, C++, NumPy/Pandas, Slurm-based HPC (CPU and GPU), Git
Languages	Chinese (native), English (professional)

TEACHING & LEADERSHIP

Learning Assistant, Classical Mechanics and Electromagnetism **Jan. 2025 – Present**

Department of Physics and Astronomy *University of California, Irvine*

- Lead weekly discussion and lab sections; coach students in experimental design and uncertainty analysis.

Webmaster, Society of Physics Students at UCI **2026 – Present**

Executive Board · Maintain the chapter website and publish events and outreach materials *University of California, Irvine*

HONORS & AWARDS

- Caltech Summer Undergraduate Research Fellowship (SURF), 2026, $8,110 award.
- UCI Undergraduate Research Opportunities Program (UROP) Award, 2024 and 2025.
- Dean's Honor List, University of California, Irvine, six quarters (2024–2026).